AP CALCULUS AB - UNIT 1

1.3 Estimating Limit Values from Graphs

Lesson Overview

- Trace a graph toward a target from each side before reading any point marker.
- Estimate one-sided and two-sided limits from graphical evidence.
- Distinguish holes, jumps, vertical asymptotes, endpoints, corners, and oscillation.
- State the precision supported by the scale and viewing window.

Key Knowledge and Vocabulary

Trace, do not land. Follow the curve toward the target; read a filled point only when asked for the function value.

Left/right evidence. A two-sided limit requires the same vertical destination from both directions.

Endpoint rule. At a domain endpoint, only the admissible one-sided limit is relevant.

Unbounded behavior. Record the sign of each branch using $+\infty$ or $-\infty$.

Graph precision. A graph supports an estimate, not arbitrary decimals beyond its scale.

Explanation and Strategy

Step 1. Cover the filled point with your finger and trace the curve from the left and right.
Step 2. Write the two one-sided limits before deciding the two-sided limit.
Step 3. For a vertical asymptote, inspect branch direction rather than merely naming the vertical line.

Solved Examples

Worked Example: Hole with a different point value

In Graph A, determine $\lim_{x\to 1} f(x)$ and $f(1)$.

Solution. The curve approaches the open point $(1, 1)$ from both sides, so the limit is 1. The filled point is at $(1, 4)$, so $f(1) = 4$.

Worked Example: Jump behavior

In Graph B, determine the left-hand and right-hand limits at $x = 2$.

Solution. The left branch approaches 3.5 and the right branch approaches 5. Since they differ, the two-sided limit does not exist. The filled point gives $f(2) = 1$ but does not affect the limit.

Worked Example: Vertical asymptote signs

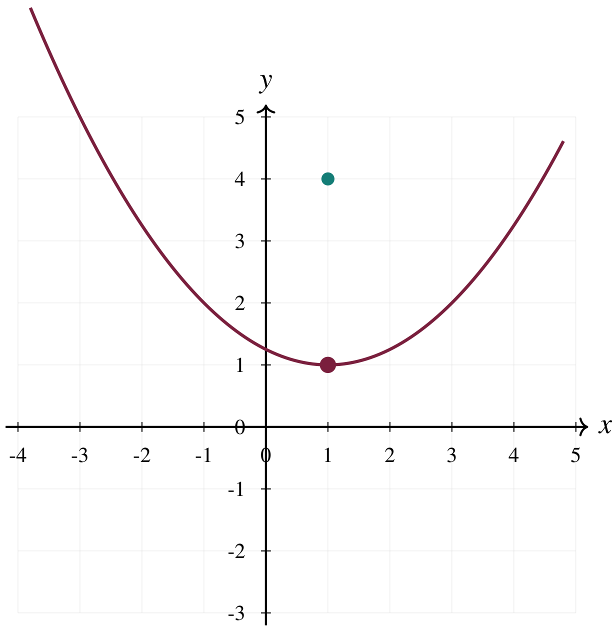
In Graph C, describe the behavior at $x = 1$.

Solution. From the left, the branch decreases without bound: $\lim_{x\to 1^-} f(x) = -\infty$. From the right, it increases without bound: $\lim_{x\to 1^+} f(x) = +\infty$.

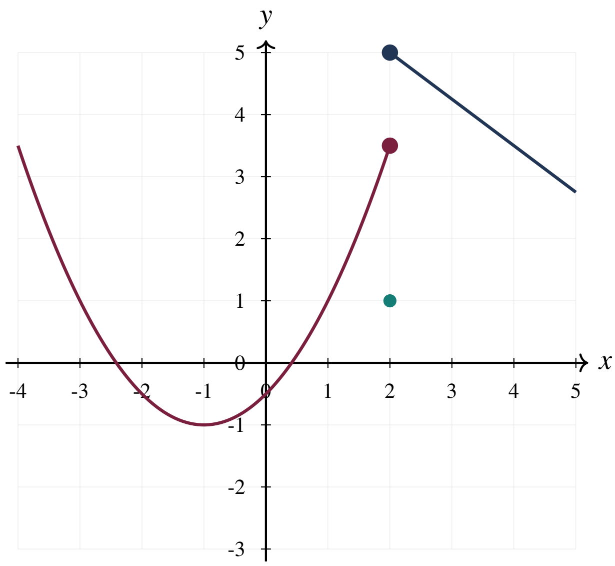
Leveled Practice

Shared Representation / Data

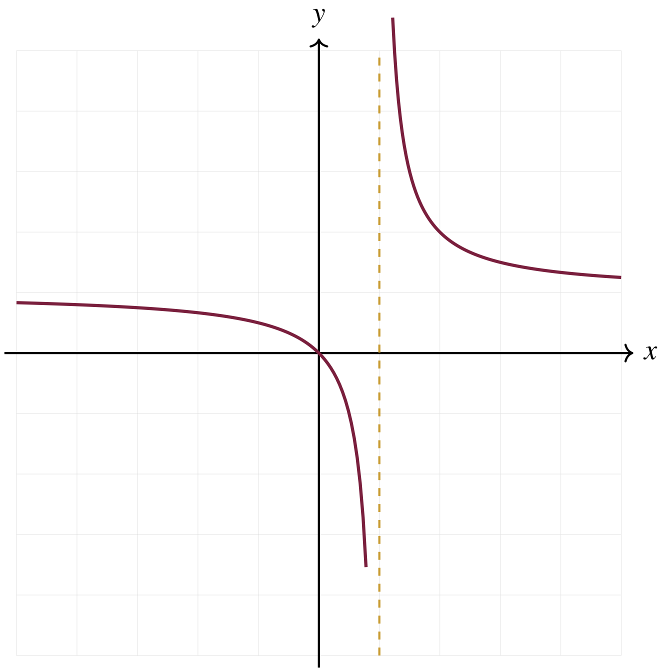
Graph A



Graph B



Graph C



Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Using Graph A, find $\lim_{x\to 1^-} f(x)$.

2. Using Graph A, find $f(1)$.

3. Using Graph B, find $f(2)$.

4. Using Graph C, identify the vertical asymptote.

Medium Level

Apply the idea in one or two connected steps.

5. Using Graph A, explain why changing the filled point would not change the limit at $x = 1$.

6. Using Graph B, state $\lim_{x\to 2^-} f(x)$ and $\lim_{x\to 2^+} f(x)$.

7. Using Graph C, write both one-sided infinite-limit statements.

8. A curve has a sharp corner at $(3, -2)$ but approaches -2 from both sides. What is the limit at $x = 3$?

Hard Level

Connect representations, conditions, and justification.

9. A graphing window shows a curve approaching about 2.4 near $x = 5$, with vertical grid spacing 1. Which is more defensible: 2.4 or 2.41763? Explain.

10. A graph is defined only for $x \geq -4$ and begins at $(-4, 7)$. State the appropriate limit at the endpoint.

11. A graph approaches 0 while oscillating with shrinking amplitude near $x = 0$. What limit is suggested?

12. A graph oscillates between -1 and 1 with no shrinking amplitude near $x = 0$. What is the two-sided limit?

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Construct a verbal description of a graph satisfying $\lim_{x\to 2^-} f(x) = -1$, $\lim_{x\to 2^+} f(x) = 4$, and $f(2) = 0$.

14. A graph appears continuous at coarse scale but reveals a hole when zoomed in. Explain why the first window was insufficient evidence.

15. Two graphs agree on every punctured neighborhood of $x = 3$ but have different point values at 3. Compare their limits at 3.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Draw a graph with $f(0) = 2$, $\lim_{x\to 0} f(x) = -1$, and a vertical asymptote at $x = 3$ with both sides tending to $+\infty$. Label all special points.

17. Can a graph cross a horizontal line $y = L$ infinitely many times near $x = c$ and still satisfy $\lim_{x\to c} f(x) = L$? Explain.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. At $x = a$, a graph approaches 6 from both sides but has a filled point at height -3. Which statement is true?

(A) $\lim_{x \to a} f(x) = -3$

(B) $\lim_{x \to a} f(x) = 6$

(C) The limit DNE.

(D) $f(a) = 6$

2. Which graphical feature necessarily prevents a finite two-sided limit at the target?

(A) A corner

(B) A cusp whose sides meet

(C) A jump with unequal one-sided heights

(D) A hole

3. If the left branch tends to $-\infty$ and the right branch tends to $+\infty$ at $x = 4$, then

(A) $\lim_{x \to 4} f(x) = 0$

(B) $\lim_{x \to 4} f(x) = +\infty$

(C) $\lim_{x \to 4} f(x)$ DNE

(D) $f(4)$ must be undefined

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator graph suggests a limit of 1.732 at $x = 2$, but the curve is steep and the window uses y-scale 10. What is the best next step?

(A) Report 1.732000 immediately.

(B) Zoom and confirm with a numerical table from both sides.

(C) Read the filled point only.

(D) Conclude DNE.

5. A graphing calculator displays $f(x) = \frac{x^2-9}{x-3}$ as a line near $x = 3$. Which limit is suggested?

(A) 0

(B) 3

(C) 6

(D) DNE

Free-Response Practice – No Calculator

Use the shared Graphs A–C.

(a) For Graph A, state the limit and point value at $x = 1$. [2 points]

(b) For Graph B, explain why the two-sided limit at $x = 2$ does not exist. [2 points]

(c) For Graph C, state both one-sided limits at the vertical asymptote. [2 points]

Free-Response Practice – Calculator Required

A graphing calculator is used to inspect $F(x) = \dfrac{\sin(40x)}{20x}$ near $x = 0$.

(a) Use a suitable window or table to estimate $\lim_{x \to 0} F(x)$. [2 points]

(b) Explain why a wide window might make the limit difficult to see. [2 points]

(c) State one numerical check that would strengthen the graphical conclusion. [2 points]